



**Karolinska
Institutet**

Master's Programme in Biostatistics and Data Science
Autumn Semester 2025
Biostatistics 1: Introduction to Biostatistics, 5BD000, 7.5 Credits

Assignment 1

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Within this assignment, data from the [swedish cancer registry](#) and [statistics database](#) will be used. The focus is set on new colon cancer cases per 5-year age groups from 1970 until 2020. The purpose is to describe and analyze patterns across the 50 year time span.

1 Number of Colon Cancer Cases by Age Group and Sex

1.1 Loading and Describing the Cases Data

We start by loading the data from *cases-1.tsv*.

```
cases <- read.delim("cases-1.tsv", sep = "\t", header = TRUE)

cases$year <- as.numeric(as.character(cases$year))

cases$agegroup <- factor(cases$agegroup, levels = unique(cases$agegroup))
```

Figure 1: R Code for loading the cases data

In Table 1 is the the summary of the data displayed.

agegroup	year	sex	n
0-4 : 106	Min. :1970	Length:1908	Min. : 0.00
5-9 : 106	1st Qu.:1983	Class :character	1st Qu.: 4.00
10-14 : 106	Median :1996	Mode :character	Median : 28.00
15-19 : 106	Mean :1996		Mean : 96.73
20-24 : 106	3rd Qu.:2009		3rd Qu.:177.25
25-29 : 106	Max. :2022		Max. :621.00
(Other):1272			

Table 1: Summary of Cases Table

Table 2 shows the head (first six rows) of the cases data.

	agegroup	year	sex	n
1	0-4	2022.00	Male	0
2	5-9	2022.00	Male	0
3	10-14	2022.00	Male	1
4	15-19	2022.00	Male	8
5	20-24	2022.00	Male	5
6	25-29	2022.00	Male	4

Table 2: Snippet of Cases Table

1.2 Plotting Number of Cases by Age Group and Sex

Here, we sum up the number of cases for each age group and sex. This is done by first grouping by agegroup and sex and then summing up the number of cases.

```
# Sum of cases per year and sex
cases_agg_total <- aggregate(n ~ agegroup + sex, data = cases, FUN = sum)
```

Figure 2: R Code for Summing the Cases per Year and Sex

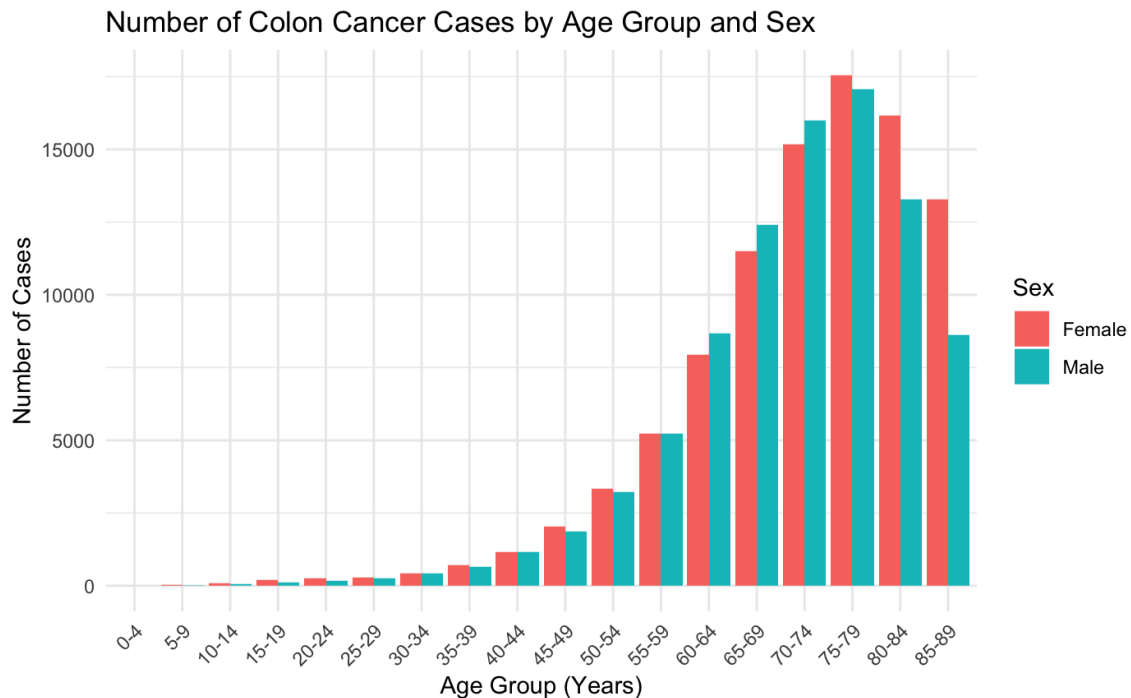


Figure 3: Number of Colon Cancer Cases by Age Group and Sex

1.3 Observations

In Figure 3, it can be seen that age has a large effect on the number of cases.

The graph demonstrates an extremely strong positive correlation between age and the number of cases. The number of cases begin to increase rapidly starting around the 45-49 age group. The vast majority of cases occur in the elderly population, with the peak incidence observed in the 75-79 age group. We see that the number of cases drop rapidly at around ages 75 – 79. This is likely because the population in this age range is smaller.

2 Total Number of Cases in each Calendar Year

2.1 Plotting Number Cases by Year

Here, we plot the total cases per calendar year. This is done by grouping by year and sex and then summing up the total cases.

```
cases_yearly <- aggregate(n ~ year + sex, data = cases, FUN = sum)
```

Figure 4: R Code for Summing the Cases per Year and Sex

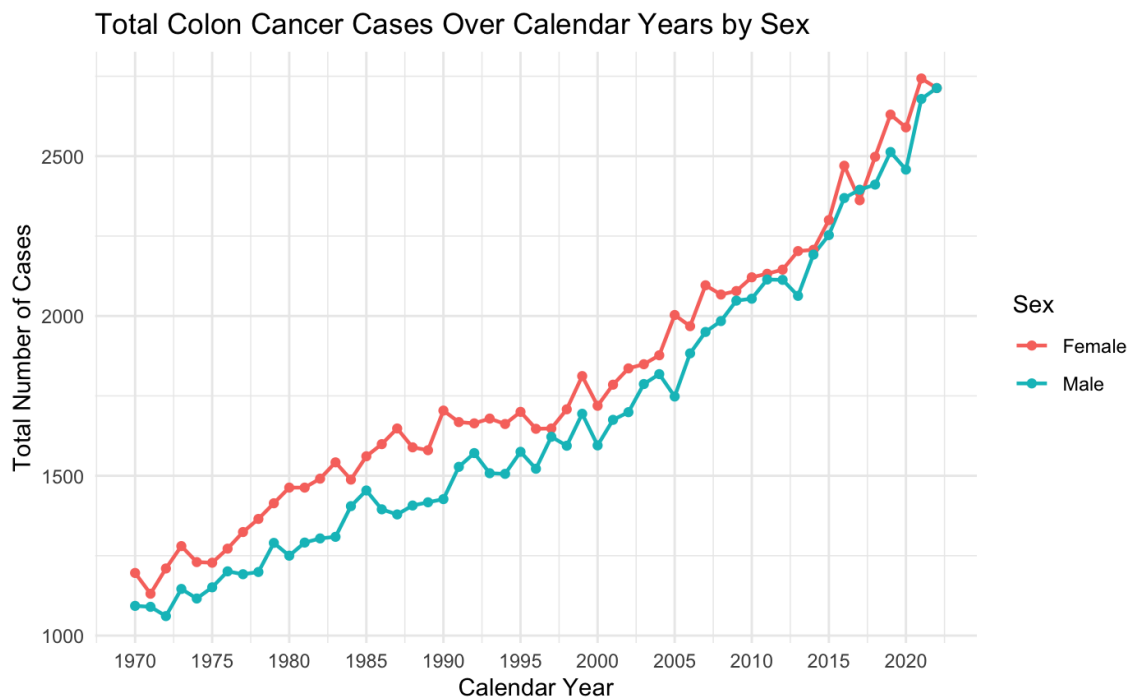


Figure 5: Total Number of Cases in each Calendar Year by Males and Females

2.2 Observations

A clear upwards trend in the total number of colon cancer cases for both males and females can be observed. In most years throughout the period, the total number of cases is slightly higher for females (orange) than for males (blue). In the most recent years, the case counts for males and females seem to converge, especially in comparison to the 1980 to 1990 period, indicating a more equal distribution between the sexes.

3 Loading and Plotting the Population Data

3.1 loading Data

The population data is loaded from *population-1.tsv* file.

```
population <- read.delim("population-1.tsv", sep = "\t", header = TRUE)
population$year <- as.numeric(as.character(population$year))
population$agegroup <- factor(population$agegroup, levels = unique(population$agegroup))
```

Figure 6: R Code for loading the population-1.tsv

In Table 3 is the the summary of the data displayed.

agegroup	sex	year	n_pop
0-4 : 106	Length:1908	Min. :1970	Min. : 26513
5-9 : 106	Class :character	1st Qu.:1983	1st Qu.:215564
10-14 : 106	Mode :character	Median :1996	Median :269226
15-19 : 106		Mean :1996	Mean :248347
20-24 : 106		3rd Qu.:2009	3rd Qu.:299042
25-29 : 106		Max. :2022	Max. :400939
(Other):1272			

Table 3: Summary of the Population Table

Table 4 shows the head (first six rows) of the cases data.

	agegroup	sex	year	n_pop
1	0-4	Male	1970.00	296143
2	5-9	Male	1970.00	295092
3	10-14	Male	1970.00	272559
4	15-19	Male	1970.00	281997
5	20-24	Male	1970.00	336648
6	25-29	Male	1970.00	329316

Table 4: Snippet of the Population Table

This table contains information on population sizes in certain age groups over the years, seperately for males and females. There are 106 observation in each age group, one for each sex and year.

3.2 Plotting Data

Below is a plot of the population size, by sex, per calendar year.



Figure 7: Line Chart of Population by Year

This plot in Figure 7 shows the yearly population size for each 5-year age group, separated by sex. The dense overlapping of the lines make the graphic complex and hard to read.

To simplify this, we can put two agegroups together (in a new 10 year interval).

```
merged_data
population$agegroup2 <- ifelse(population$agegroup %in% c("0-4", "5-9"), "0-9",
  ifelse(population$agegroup %in% c("10-14", "15-19"), "10-19",
    ifelse(population$agegroup %in% c("20-24", "25-29"), "20-29",
      ifelse(population$agegroup %in% c("30-34", "35-39"), "30-39",
        ifelse(population$agegroup %in% c("40-44", "45-49"), "40-49",
          ifelse(population$agegroup %in% c("50-54", "55-59"), "50-59",
            ifelse(population$agegroup %in% c("60-64", "65-69"), "60-69",
              ifelse(population$agegroup %in% c("70-74", "75-79"), "70-79",
                ifelse(population$agegroup %in% c("80-84", "85-89"), "80-89",
                  NA)))))))))

pop_collapsed <- aggregate(
  n_pop ~ year + sex + agegroup2,
  data = population,
  FUN = sum
)
```

Figure 8: R Code for combining Age Groups

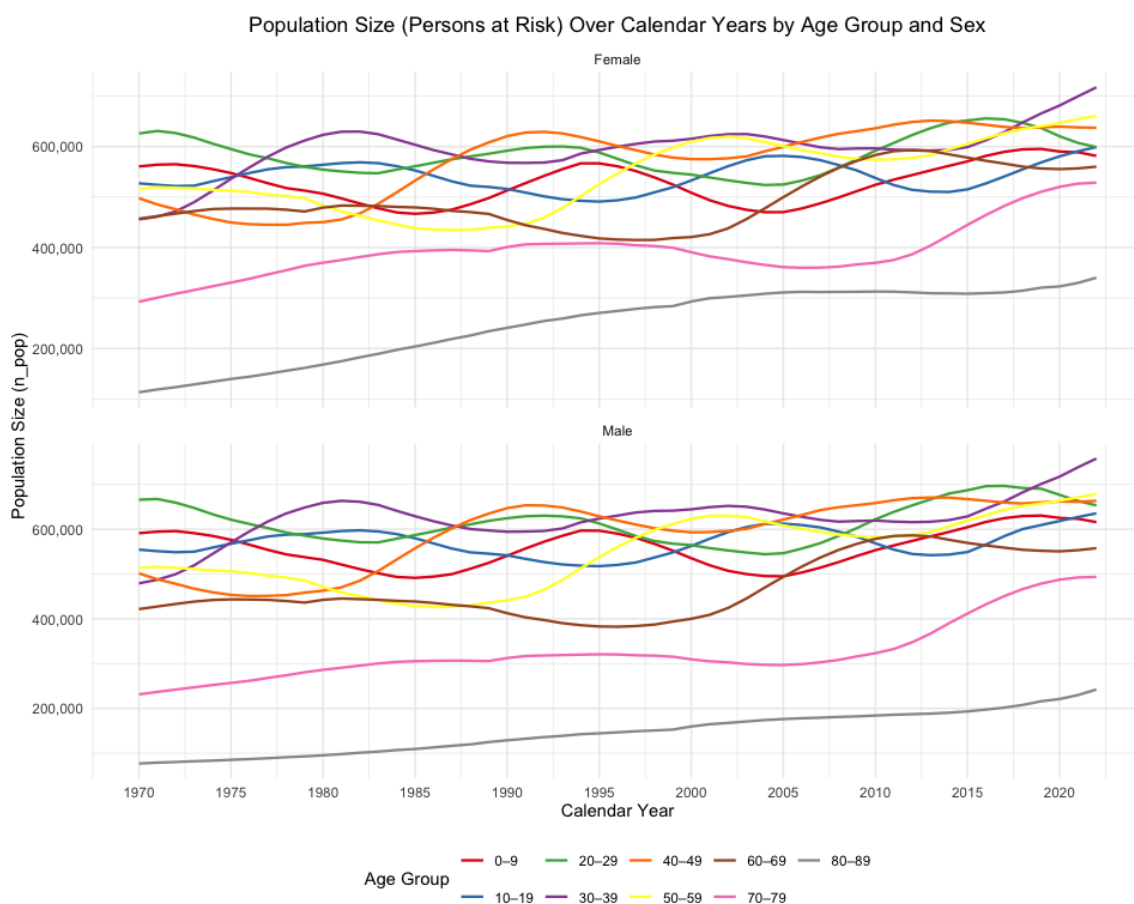


Figure 9: Line Chart of Population by Year, with Two Age Groups Grouped

This grouped version in Figure 9 now combines two age groups into a 10-year interval, making it easier to compare.

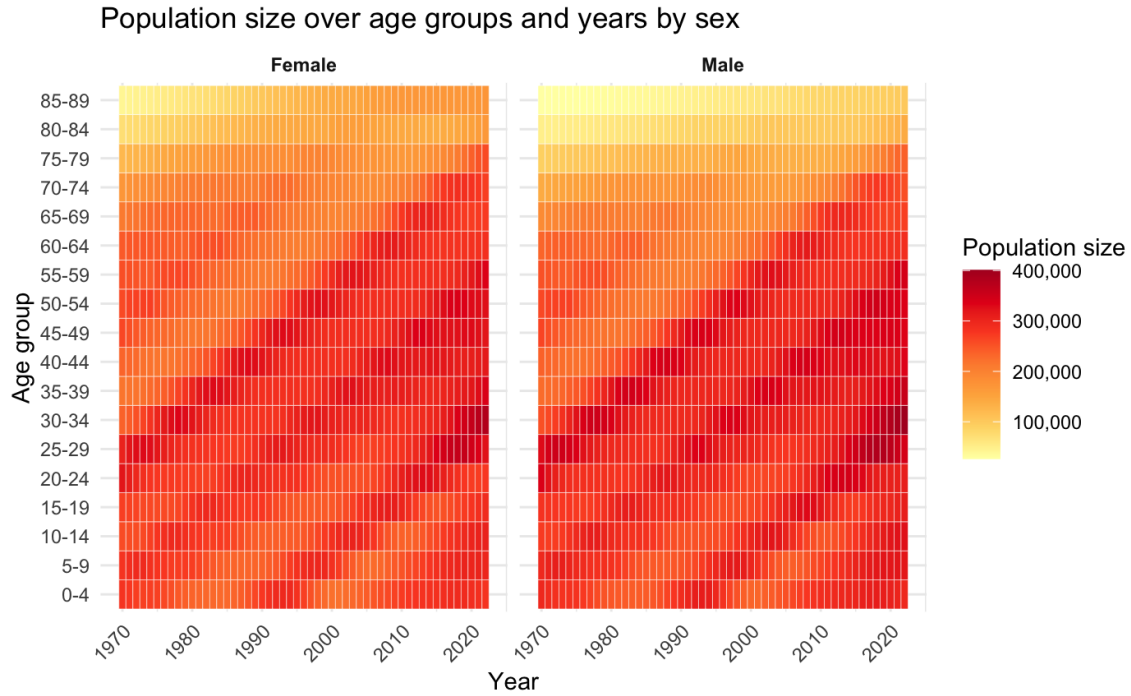


Figure 10: Heatmap of Population Over Age Groups

The heatmap in Figure 10 shows the population size across age groups (rows) and years (columns) for each sex. Color intensity highlights demographic shifts over time.

3.3 Observations

The population shows a cyclic behavior with an interval of approximately 20–30 years. This probably happens because people tend to have children at this age, thus resulting in a new large emerging age group being born, that repeats the cycle.

We can also see how the same large group of people moves through the age groups as time goes forward. If an age group is big in one year, it usually stays big when it becomes the next age group later.

Strong age groups stay strong, and small age groups stay small. This shows how generations stay prevalent as they age.

4 Merging Tables

Now, we merge the two tables, and make sure that they match.

```
merged_data <- merge(cases, population, by = c("agegroup", "year", "sex"), all = TRUE)

check_merge <- merged_data[is.na(merged_data$n) | is.na(merged_data$n_pop), ]

cat("\n--- Data Alignment Check ---\n")
if (nrow(check_merge) == 0) {
  cat("The 'cases' and 'population' files include the same combinations.\n")
} else {
  cat("The 'cases' and 'population' files do NOT include the same combinations.")
  cat("Missing values found:\n")
  cat("Number of combinations with missing data:", nrow(check_merge), "\n")
  print(check_merge)
}

totals_yearly <- aggregate(
  cbind(total_cases = n, total_population = n_pop) ~ year + sex,
  data = complete_merged_data,
  FUN = sum
)

--- Data Alignment Check ---
The 'cases' and 'population' files include the same combinations.
```

Figure 11: R Code for Merging and Validating the Tables, along with output

The check shows that both tables have the same years, age groups, and sex categories. Because the data match perfectly, we can merge the tables without any problems.

	year	sex	total_cases	total_population
1	1970.00	Female	1196	4045318
2	1971.00	Female	1131	4066592
3	1972.00	Female	1210	4077814
4	1973.00	Female	1280	4089903
5	1974.00	Female	1230	4108883
6	1975.00	Female	1228	4127179

Table 5: Snippet of the Merged Table

Table 5 shows the merged data, now including both cases and population.

5 Calculating the Incidence Rate

In epidemiology, an *incidence rate* describes how frequently new cases of a disease occur in a population over a given time period. In continuous time, it is usually defined as

$$IR = \frac{NC}{PT}.$$

where NC is the number of new cases during the period, and PT is the person-time at risk during the period.

In this assignment, we approximate the incidence rate by dividing the number of new colon cancer cases by the population size:

$$IR_{a,s,t} = \frac{C_{a,s,t}}{P_{a,s,t}} \times 100,000,$$

where C denotes the number of cases, P , the population, a denotes age group, s sex, and t calendar year.

When we sum over all age groups we obtain the *crude incidence rate*

$$CIR_{s,t} = \frac{\sum_a C_{a,s,t}}{\sum_a P_{a,s,t}} \times 100,000.$$

This method ignores the exact person-time at risk and treats the yearly population size as if everyone contributes one year of follow-up.

Strictly speaking, it therefore estimates an incidence *proportion* rather than a true incidence rate. However, with large populations and annual data this is a common and reasonable approximation, as long as the population size does not change dramatically within each year.

A limitation is that the crude rate does not adjust for changes in the age distribution over time, which is why the age-specific rates are also important to inspect.

```
# --- Version 1. Age-Specific Incidence Rate (ASIR) ---
merged_data$asir <- merged_data$n / merged_data$n_pop * 100000

# --- Version 2. Crude Incidence Rate (CIR) ---
totals_yearly$cir <- totals_yearly$total_cases / totals_yearly$total_population * 100000
```

Figure 12: R Code for Calculating the Incidence Rates

	agegroup	year	sex	n_cases	n_pop	asir
1903	85-89	2020.00	Female	412	167759	245.59
1904	85-89	2020.00	Male	269	94931	283.36
1905	85-89	2021.00	Female	498	170428	292.21
1906	85-89	2021.00	Male	349	98195	355.42
1907	85-89	2022.00	Female	451	172566	261.35
1908	85-89	2022.00	Male	304	101450	299.66

Table 6: the tail of the merged data table, with the age-specific incidence rate

	year	sex	total_cases	total_population	cir
1	1970.00	Female	1196	4045318	29.57
2	1971.00	Female	1131	4066592	27.81
3	1972.00	Female	1210	4077814	29.67
4	1973.00	Female	1280	4089903	31.30
5	1974.00	Female	1230	4108883	29.94
6	1975.00	Female	1228	4127179	29.75

Table 7: the head of the totals_yearly table with the crude incidence rate

6 Smoothed Incidence Rate of Colon Cancer over Calendar Time

6.1 Applying a Smoother

A smoother (geom_smooth from ggplot2 package) is applied to smooth the data.

```
# Applying a smoother (LOESS)
geom_smooth(method = "loess", se = FALSE, linewidth = 0.8) +
```

Figure 13: R Code for smoothing the incidence rate

6.2 Plotting the Data

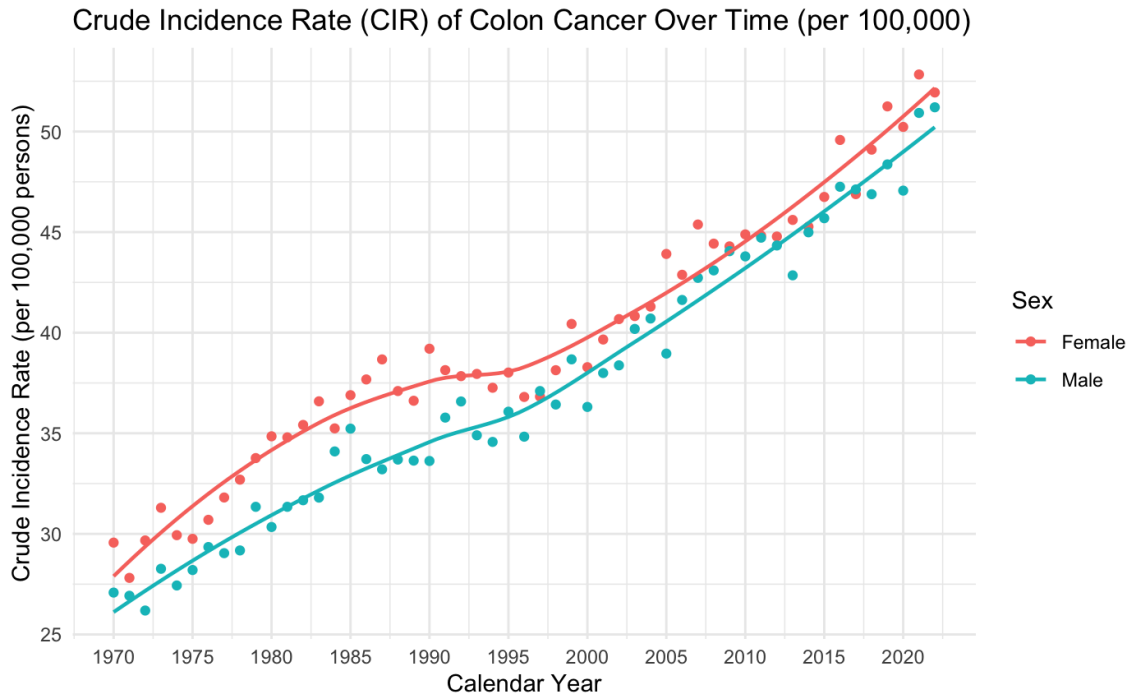


Figure 14: Crude Incidence Rate (CIR) of Colon Cancer Smoothed Over Time

6.3 Observations

The smoothed lines show a clear increase in colon cancer rates over the whole time period for both men and women, just as in Section 2. Women have slightly higher rates than men, but the

overall trend is similar. The smoothing also makes it easier to see a small slowdown around the 1990s for females before the rates rise again.

7 Poisson Regression

7.1 Poisson Modeling

Here we fit a Poisson model, to fit the total cases to the year and sex.

```
pois_model <- glm(
  total_cases ~ year + sex,
  offset = log(total_pop),
  family = poisson,
  data = year_sex_totals
)
summary(pois_model)
```

Figure 15: R code for Poisson regression model

Coefficient	Estimate	Std. Error	p-value
Intercept	-29.69	0.307	<0.001
Year	0.01094	0.0001536	<0.001
Male (vs Female)	-0.05592	0.00466	<0.001

Table 8: Estimated Coefficients from the Poisson Regression Model

7.2 Observations

Table 8 shows the fitted Poisson model. The coefficient for calendar year is positive, indicating a clear increasing trend in incidence over time. The sex coefficient is negative, implying that males have slightly lower incidence than females.

The year coefficient is positive, implying that the cancer rate is increasing over time.

It is worth mentioning that the p values are very small, which means the results are statistically significant.

8 Predicting Crude Incidence Rates with the Task 7 Model

8.1 Predicting

Here we use the poisson model built in Section 7.1 to predict incidence rates among males and females in 1970 and in 2020.

```
predicted_data <- data.frame(  
  year = c(1970, 1970, 2020, 2020),  
  sex = as.factor(c("Male", "Female", "Male", "Female")),  
  total_pop = 100000  
)  
  
predicted_data$cir <- predict(pois_model, newdata = predicted_data, type = "response")
```

Figure 16: R Code for Predicting Crude Incidence Rates

	year	sex	total_population	predicted_cir	sex_difference
1	1970	Male	100000	27.7375	–
2	1970	Female	100000	29.3327	1.5952
3	2020	Male	100000	47.9420	–
4	2020	Female	100000	50.6992	2.7572

Table 9: The Predicted_data1 Table

8.2 Assumptions

1. Regarding calendar year: Since only crude incidence rates are used here, we assume that differences in age structure across calendar years do not substantially affect the incidence rates. Thus, any observed changes are interpreted as reflecting true disease occurrence.
2. Regarding sex: Also, we assume that the age distribution of males and females are rather similar, so that the effect of age composition between sexes can be neglected. Consequently, the observed differences in crude rates between males and females are interpreted as the intrinsic effect of sex on disease.
3. This model also assumes that the crude rate increases linearly with age, and that the rate of change is the same for both sexes.

9 Incidence Smoothing by Age and Sex

The following figure displays the age-specific incidence rates calculated in Section 5. The data is grouped by sex and age group. To simplify the graphs against yearly fluctuations, a smooth trend line (LOESS method) is used, making it easier to interpret.

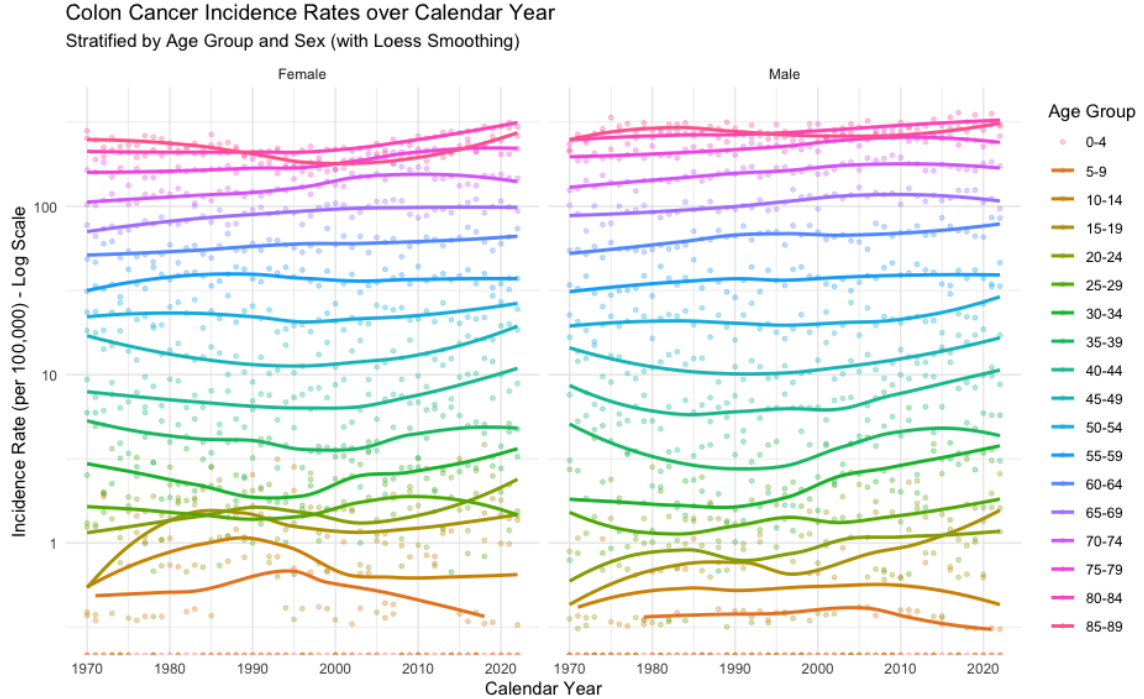


Figure 17: Trends in Colon Cancer Incidence Rates by Age Group and Sex.

9.1 Observations

The Figure 17 shows the incidence rate data from 1970 to 2020. We can observe that the age effect is highly significant, with incidence rates increasing exponentially with age (represented by the clear vertical separation on the log scale), indicating that age is the strongest risk predictor. There is a general upward trend in incidence rates across most adult age groups during the study period. Furthermore, the data indicates a sex difference, as males consistently show slightly higher incidence rates than females of the same age, although the temporal trends (slopes of the lines) are similar between them. Finally, the smoothed curves exhibit some non-linearity, meaning the change in incidence rates over calendar years is not strictly linear, which justifies the use of splines in the model.

10 Fitting an Age-adjusted Poisson Model

10.1 fitting the model

Here we fit a spline to the merged data, for each sex and age group. This is then used to compare the incidence rates in 1970 and 2020 for males and females.

```
model_interaction <- glm(n ~ ns(year, df = 4)*sex + ns(year, df = 4)*agegroup,
                        offset = log(n_pop),
                        family = poisson(link = "log"),
                        data = merged_data)

target_data <- data.frame(
  year = c(1970, 1970, 2020, 2020),
  sex = as.factor(c("Male", "Female", "Male", "Female")),
  agegroup = "70-74",
  n_pop = 100000
)

target_data$predicted_rate <- predict(model_interaction,
                                     newdata = target_data,
                                     type = "response")
```

Figure 18: R Code for predicting Incidence Rates

	year	sex	agegroup	n_pop	predicted_rate
1	1970.00	Male	70-74	100000.00	124.08
2	1970.00	Female	70-74	100000.00	110.90
3	2020.00	Male	70-74	100000.00	165.98
4	2020.00	Female	70-74	100000.00	148.43

Table 10: The Target_data Table with the Predicted Incidence Rate

This models uses splines to capture the non linearity of the effect of the year, and contains interactions between the year and sex, and the year and age group.

10.2 observations

A rather surprising result here is that now, men have a higher predicted rate than women, while in earlier models, where we did not account for the age group, the rate was higher for women. A possible reason for this is that the female population has more individuals in the older age groups, where colon cancer is more common.

10.3 exploring the incidence rates across age groups

To further explore this, difference, we can plot the incidence rate by age groups.

```
age_levels <- unique(merged_data$agegroup)

target_data <- expand.grid(
  year = seq(1970, 2020, by=1),
  sex = c("Male", "Female"),
  agegroup = age_levels,
  n_pop = 100000
)

target_data$agegroup <- factor(target_data$agegroup, levels = age_levels)
target_data$sex <- factor(target_data$sex, levels = c("Male", "Female"))

target_data$predicted_rate <- predict(model_interaction,
  newdata = target_data,
  type = "response")

target_data <- target_data %>% group_by(agegroup, sex) %>% summarize(predicted_rate = mean(predicted_rate))
```

Figure 19: R Code for calculating incidence rates per age group

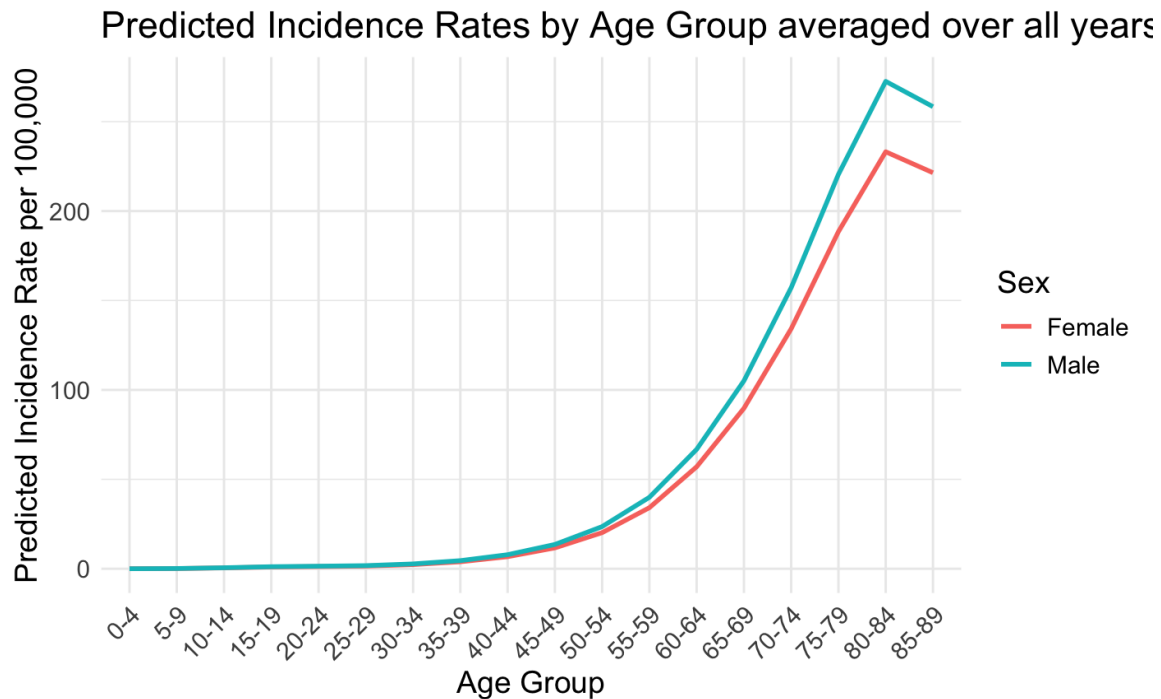


Figure 20: Cases Among the Population

As we can see here, the colon cancer incidence rate is much higher in older age groups, and from fig. 10 we can see that women tend to reach older ages, explaining why women have a higher incidence rate overall, but when controlling for age groups, men can have a higher rate.

11 Refitting the Model Using Splines for the Effect of Calendar Year and Age Group (Using the Mid Point of Each Age Group)

Using splines for the effect of calendar year and age group. This is done by including an interaction between the year and age_mid, as well as year and sex.

```
df_q11 <- merged_data %>%
  mutate(
    lower_bound = as.numeric(str_extract(agegroup, "[0-9]+")),
    age_mid = lower_bound + 2
  )
model_spline_age <- glm(
  formula= n ~ ns(year, df = 4)*ns(age_mid, df=4) + ns(year, df = 4)*sex,
  offset=log(n_pop),
  family = poisson(link = "log"),
  data = df_q11)

pred_data <- expand.grid(
  year = 1970:2020,
  sex = df_q11$sex,
  age_mid = c(52, 72, 87),
  n_pop = 100000
)

pred_data$fitted_rate <- predict(model_spline_age, newdata = pred_data, type = "response")

obs_data <- df_q11 %>%
  filter(agegroup %in% c("50-54", "70-74", "85-89")) %>%
  mutate(
    obs_rate = (n / n_pop) * 100000,
    target_age = case_when(
      agegroup == "50-54" ~ 52,
      agegroup == "70-74" ~ 72,
      agegroup == "85-89" ~ 87
    )
  )
```

Figure 21: R Code for extracting age groups and refitting the model using them

Colon Cancer Incidence: Model vs Observed

Lines = Spline Model Predictions (Ages 52, 72, 87)

Points = Observed Crude Rates (Groups 50-54, 70-74, 85-89)

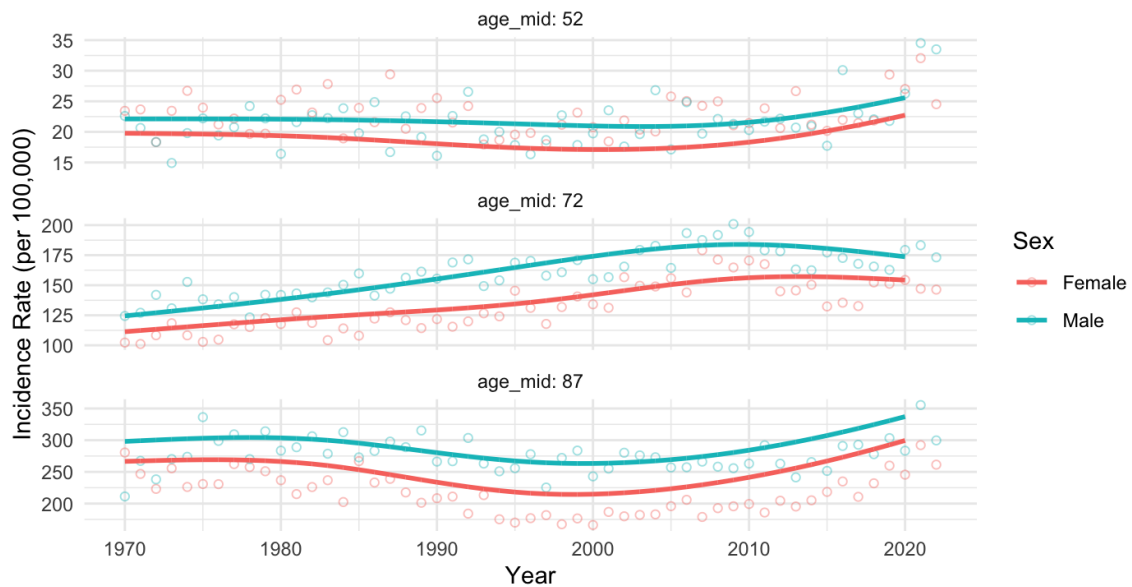


Figure 22: Colon Cancer Incidences, Splines Model Compared to Observed

11.1 Observations

We see that the incidence rate is higher for men than for women, especially in older age groups. In the 52 year old age group, the difference is not large.

In the age groups 70-74 and 85-89, the observed crude rate and the predicted spline model rate generally maintain the same relative position and trend between sexes. Conversely, in the 50-54 age group, females exhibited a higher crude incidence rate in most years, but the spline model predicted a higher incidence rate for males.

12 Age Standardization

In this section, we aim to find the age-standardized incidence rates across calendar years. This is done by applying each year's age-specific incidence rates to the standard population, which is the 2022 population.

```
std_pop_2022 <- df_q11 %>%
  filter(year == 2022) %>%
  select(sex, age_mid, std_pop = n_pop)

df_standardized <- df_q11 %>%
  mutate(obs_rate_per_person = n / n_pop) %>%
  left_join(std_pop_2022, by = c("sex", "age_mid")) %>%
  group_by(year, sex) %>%
  summarise(
    expected_cases = sum(obs_rate_per_person * std_pop),
    total_std_pop = sum(std_pop),
    .groups = "drop"
  ) %>%
  mutate(
    rate = (expected_cases / total_std_pop) * 100000,
    type = "Age-Standardized (Observed)"
  ) %>%
  select(year, sex, rate, type)

df_crude <- df_q11 %>%
  group_by(year, sex) %>%
  summarise(
    total_cases = sum(n),
    total_pop = sum(n_pop),
    .groups = "drop"
  ) %>%
  mutate(
    rate = (total_cases / total_pop) * 100000,
    type = "Crude (Non-standardized)"
  ) %>%
  select(year, sex, rate, type)

df_final_plot <- bind_rows(df_standardized, df_crude)
```

Figure 23: R Code for Age Standardization

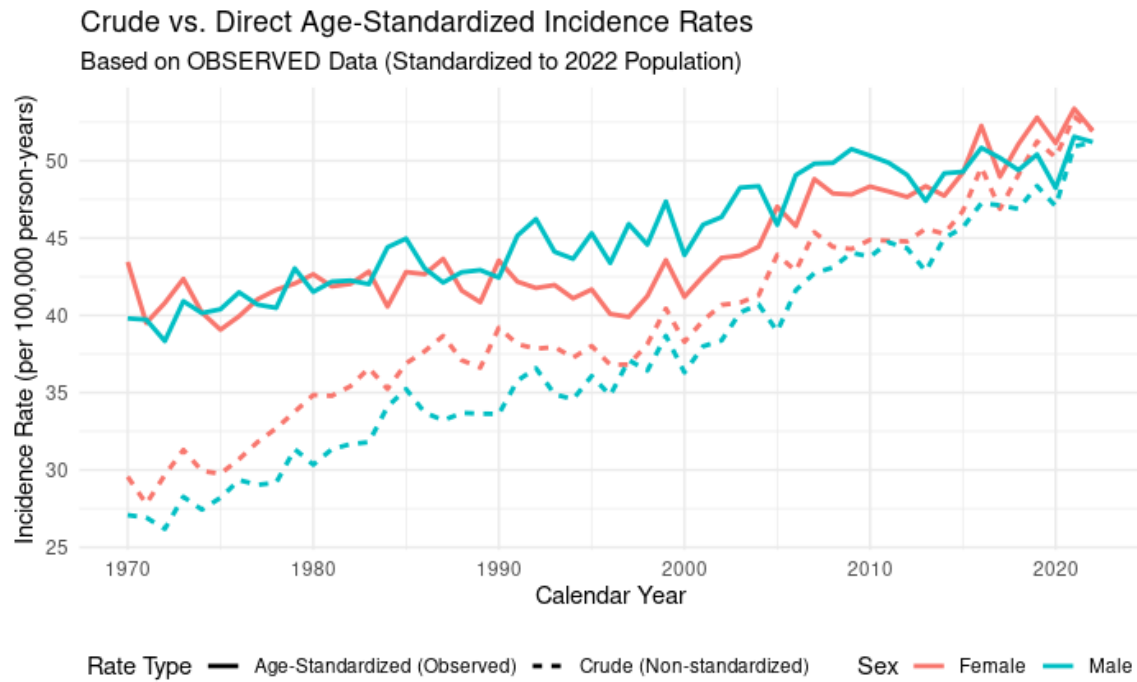


Figure 24: Age Standardized Incidence Rates

12.1 Observations

Here we can see the age-standardized incidence rate, compared to the observed incidence rate. We see that even after standardizing for population age, there is an upward trend in colon-cancer rates, both for males and females.

13 Age-Standardized Rates Based on the Regression Model including Age, Year and Sex

Instead of standardizing the observed rates, the standardization is applied to the predicted rates from the model.

```
std_pop_weights <- df_q11 %>%
  filter(year == 2022) %>%
  select(sex, age_mid, std_weight = n_pop)

df_asr_observed <- df_q11 %>%
  arrange(sex) %>%
  mutate(raw_rate = n / n_pop) %>%
  left_join(std_pop_weights, by = c("sex", "age_mid")) %>%
  group_by(year, sex) %>%
  summarise(
    weighted_sum = sum(raw_rate * std_weight),
    total_weight = sum(std_weight),
    .groups = "drop"
  ) %>%
  mutate(
    rate = (weighted_sum / total_weight) * 100000,
    method = "Observed (Direct)"
  )

pred_grid <- expand_grid(
  year = 1970:2022,
  age_mid = unique(df_q11$age_mid),
  sex = unique(df_q11$sex),
  n_pop = 100000
)

pred_grid$pred_count <- predict(model_spline_age, newdata = pred_grid, type = "response")
pred_grid$pred_rate_per_person <- pred_grid$pred_count / 100000
df_asr_model <- pred_grid %>%
  left_join(std_pop_weights, by = c("sex", "age_mid")) %>%
  group_by(year, sex) %>%
  summarise(
    weighted_sum = sum(pred_rate_per_person * std_weight),
    total_weight = sum(std_weight),
    .groups = "drop"
  ) %>%
  mutate(
    rate = (weighted_sum / total_weight) * 100000,
    method = "Model-Based (Predicted)"
  )
```

Figure 25: R Code for Refitting the Model

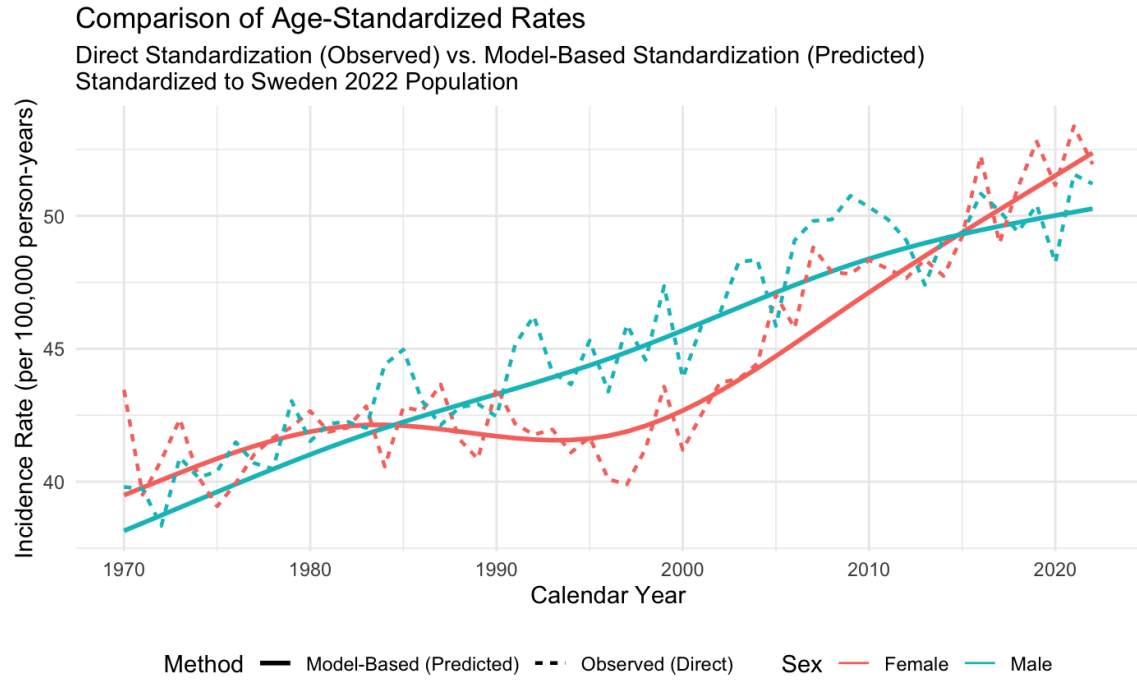


Figure 26: Age Standardized Incidence Rates

13.1 Observations

The standardized model results fit the observed data well. It shows the same trend as in Section 12.

14 Conclusions

There is clear increasing trend in colon cancer among all age groups, for both males and females. This increase is also present when accounting for the population age.

We also noticed that in general, women have a higher incidence rate of colon cancer, but when accounting for the age groups, men have a higher rate. We concluded that this is because women tend to reach older ages, where the incidence rate is higher.